Solution STA510 Exam spring 2024

Problem 1

We have the probability density function

$$f(x) = k(x-2)^2, \quad 0 \leq x \leq 2.$$

a)

In order to be proper probability density function, we have to check $f(x) \geq 0$ and $\int_0^2 f(x)dx = 1$. The first condition is trivial as $(x-2) \geq 0$ for $0 \leq x \leq 2$. For second condition we need to find the normalization constant k,

$$\begin{aligned}
\int_0^2 f(x)dx = \int_0^2 k(x-2)^2 dx = k \cdot \left[\frac{1}{3}(x-2)^3\right]_0^2 \\
= k \cdot \left(\frac{1}{3} \cdot 0 - \frac{1}{3} \cdot (-2)^3\right) = k \cdot \frac{8}{3} = 1 \\
\iff k = \frac{3}{8}.
\end{aligned}$$

b)

Assume we have a random variable X with probability density $f(x)$. Then,

$$P(X \leq 1) = \int_0^1 \frac{3}{8}(x-2)^2 dx = \frac{1}{8}\left[(x-2)^3\right]_0^1 = -\frac{1}{8} + 1 = \frac{7}{8}.$$

.

The expectation $E(X)$ is given by

$$\begin{aligned}
E(X) &= \int_0^2 x f(x)dx = \int_0^2 \frac{3}{8}x(x-2)^2 dx = \frac{1}{8}\int_0^2 (3x^3 - 12x^2 + 12x)dx \\
&= \frac{1}{8}\left[\frac{3}{4}x^4 - \frac{12}{3}x^3 + \frac{12}{2}x^2\right]_0^2 = \frac{1}{8}(12 - 32 + 24) \\
&= \frac{4}{8} = \frac{1}{2}.
\end{aligned}$$

For $Y = \frac{X_1 + X_2 - 2}{4}$ we have

$$E(Y) = E(\frac{1}{4}(X_1 + X_2 - 2)) = \frac{1}{4}E(X_1) + \frac{1}{4}E(X_2) - \frac{1}{2} = \frac{2}{8} - \frac{1}{2} = -\frac{1}{4}.$$

Problem 2

Consider a random variable X with probability density

$$f(x) = 6x(1-x), \; 0 \leq x \leq 1.$$

a)

Write down the algorithm (with words and formulas, not code) to simulate n random variables with density $f(x)$ with the acceptance-rejection method. Specify also potential constants of the algorithm.

We have that a random variable X has a pdf $f(x)$. Now, we need to find a pdf $g(y)$ and a constant c such that

$$\frac{f(t)}{g(t)} \leq c \quad \text{for all } t \text{ where } f(t) > 0.$$

Let us choose a uniform proposal $Y \sim U(0,1)$ with density $g(y) = 1$. Then we have $\frac{f(t)}{g(t)} = f(t)$. We can find the maximum of $f(t)$ in order to find the constant c.

$$\begin{aligned} f(x) &= 6x - 6x^2 \\ f'(x) &= 6 - 12x = 0 \iff x = \frac{1}{2} \\ f''(x) &= -12 < 0 \\ f(\frac{1}{2}) = 6\frac{1}{2}(1-\frac{1}{2}) &= \frac{3}{2}. \end{aligned}$$

Thus, we choose $c = 1.5$. The algorithm consist of the following steps:

1. Generate $Y \sim U(0,1)$.
2. Generate $U \sim U(0,1)$.
3. If $u < \frac{2}{3}f(y)$ accept y, otherwise reject y and repeat from step 1.

We repeat the algorithm until we have accept.

Problem 3

a)

We want to estimate the integral of

$$\theta = \int_1^6 \frac{1}{2} e^{-x^{3/4}+2} x^{1/3} dx = \int_1^6 g(x)dx$$

with $g(x) = \frac{1}{2}e^{-x^{3/4}+2}x^{1/3}$. The code for estimation 1 is given by

```
gx <- function(x)
  exp(-x^(3/4)+2) * x^(1/3) /2

Nsim <- 10000

# Estimation 1
x1 <- runif(Nsim/2,1,6)
x2 <- 5-x1
Est1 <- 5*mean(gx(c(x1,x2)))
```

This looks like a crude MC algorithm with antithetic variables. We have the antithetic pairs (X_i, Y_i) with $X_i \sim U(1,6)$ and $Y_i = a + b - X_i$ with $a = 1$ and $b = 6$ for $i = 1, \ldots, n/2$ and $n = 10000$. In the code we have $Y_i = 5 - X_i$ (wrong) which is different to $Y_i = 7 - X_i$ (correct). Thus, estimation 1 will produce wrong results. If $Y_i = 7 - X_i$ then the algorithm is correct as

$$\hat{\theta}_{AT} = \frac{b-a}{n} \sum_{i=1}^{n/2} (g(X_i) + g(Y_i)).$$

The code for estimation 2 and 3 is given by

```
# Estimation 2
x <- rgamma(Nsim,4,1.4)
Est2 <- mean(dgamma(x,4,1.4)/(gx(x)*(x<6 & x>1)))

# Estimation 3
x <- rgamma(Nsim,4,1.4)
Est3 <- mean(gx(x)*(x<6 & x>1)/dgamma(x,4,1.4))
```

In both cases, we generate random samples from a Gamma distribution with shape parameter $\alpha = 4$ and scale parameter $\beta = 1.4$, i.e. $X_i \sim Gamma(4, 1.4)$. Then, we calculate in the R-code

$$\hat{\theta}_2 = \frac{1}{n}\sum_{i=1}^{n}\frac{f(X_i)}{g(X_i)}\mathbb{I}_{(1,6)}(X_i)$$

and

$$\hat{\theta}_3 = \frac{1}{n}\sum_{i=1}^{n}\frac{g(X_i)}{f(X_i)}\mathbb{I}_{(1,6)}(X_i)$$

where $f(x)$ is the density of Gamma distribution and $\mathbb{I}_{(1,6)}(x)$ is an indicator function. We can see that $\hat{\theta}_3 = \hat{\theta}_{IS}$ is a importance sampling estimator, and $\hat{\theta}_2$ is not a correct estimator for θ because

$$\int_1^6 g(x)dx = \int_{-\infty}^{\infty}\mathbb{I}_{(1,6)}(x)g(x)dx = \int_{-\infty}^{\infty}\mathbb{I}_{(1,6)}(x)\frac{g(x)}{f(x)}f(x)dx$$

and

$$\hat{\theta}_{IS} = \frac{1}{n}\sum_{i=1}^{n}\frac{g(X_i)}{f(X_i)}\mathbb{I}_{(1,6)}(X_i)$$

with $X_1, \ldots, X_n$ r.v. with pdf. $f(x)$.

b)

The reduction of the variance in importance sampling depends on the density function $f(x)$. The aim is to find a $f(x)$ such that the variance of the estimate gets as small as possible. If the shape of $f(x)$ is very different compared to $g(x)$ then the variance of the estimator might increase compared to a crude Monte Carlo estimator.

Problem 4

a)

Suppose the lifetimes of a system follow a probability distribution with parameter θ. Assume we observe the following lifetimes $t_i, i = 1, \ldots, 20$ (in years)

$$0.62,\ 0.61,\ 1.64,\ 0.42,\ 0.45,\ 0.98,\ 0.09,\ 0.98,\ 1.91,\ 0.13 \tag{1}$$
$$0.46,\ 0.39,\ 1.10,\ 0.55,\ 0.28,\ 0.35,\ 0.83,\ 0.37,\ 0.96,\ 1.03 \tag{2}$$

and an estimator for θ is given by

$$\hat{\theta} = \frac{n}{\sum_{i=1}^{n}(t_i + 2t_i^{3/2})} = 0.481.$$

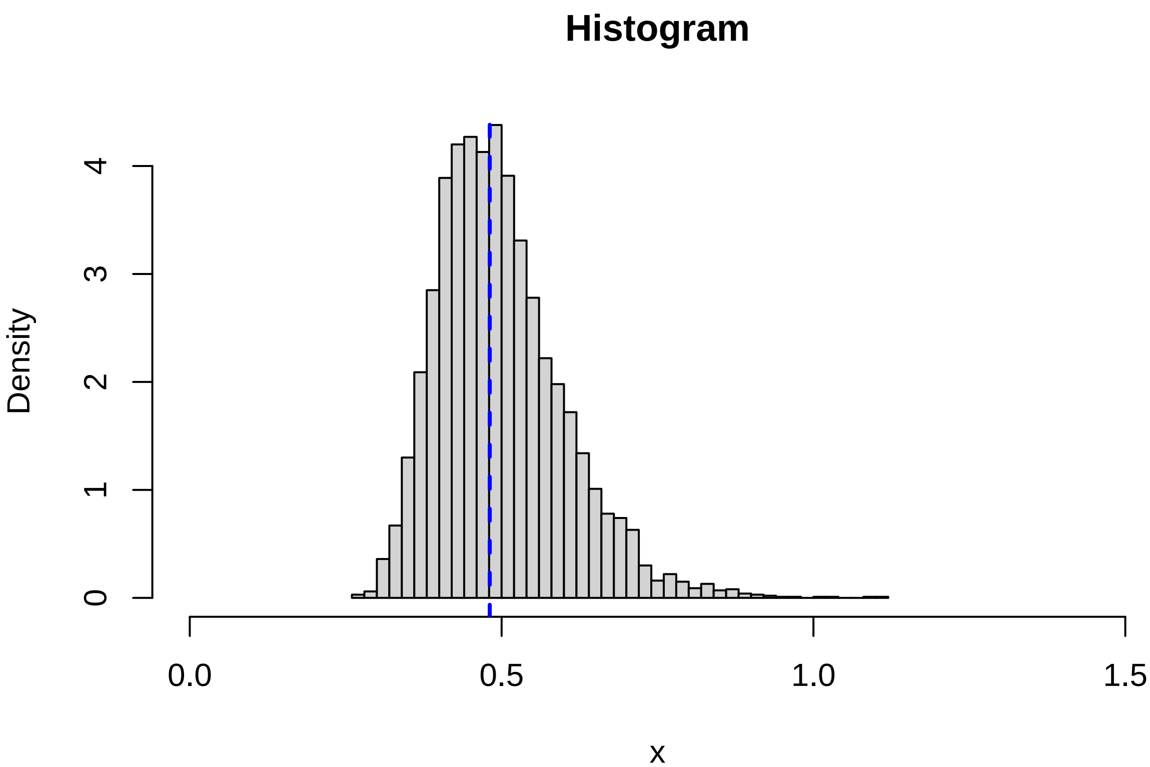
By bootstrap we create the following plot

```
LT <- c(0.62,  0.61, 1.64, 0.42, 0.45, 0.98, 0.09, 0.98, 1.91, 0.13,
        0.46,  0.39, 1.10, 0.55, 0.28, 0.35, 0.83, 0.37, 0.96, 1.03)

MLE.lifetime <- function(x){
  theta <- length(x) / sum( x + 2*x^(1.5))
  return(theta)
}
estimate <-  round(MLE.lifetime(LT),3)

## Bootstrap
set.seed(1)
B <- 5000
thetaest1B <- numeric(B)
for(i in 1:B){
  thetaest1B[i] <- MLE.lifetime(sample(LT,size=length(LT),replace=TRUE))
}

hist(thetaest1B,prob=TRUE,nclass=sqrt(B)/2, xlim=c(0,1.5),
     main="Histogram", xlab="x")
abline(v=estimate,col="blue",lwd=2,lty=2)
```



The bootstrap algorithm is:

1. Repeat step 2-3 for $b = 1, \dots, B$ where B is the number of bootstraps.
2. Simulate data $t_1^{(b)}, \dots, t_{20}^{(b)}$ from $\hat{F}$ by drawing $n = 20$ observations with replacement from the observations $t_1, \dots, t_{20}$.
3. Calculate $\hat{\theta}^{(b)}$ from $t_1^{(b)}, \dots, t_{20}^{(b)}$.

The ecdf of $\hat{\theta}^{(1)}, \dots, \hat{\theta}^{(B)}$ will be an estimate of the distribution of $\hat{\theta}$.

b)

Find the variance and the MSE of the estimator. Find a 90% standard normal bootstrap interval. Find a 90% percentile bootstrap interval.

```
# Estimate
estimate <-  round(MLE.lifetime(LT),3)

# Standard deviation
sd <-round(sd(thetaest1B),3)
sd
```

```
## [1] 0.101
```

```
# Variance
sd^2
```

```
## [1] 0.010201
```

```
# Bias
bias <- round(mean(thetaest1B),3) - estimate
bias
```

```
## [1] 0.017
```

```
# MSE
round(sd^2-bias^2,3)
```

```
## [1] 0.01
```

```
# Normal & percentile bootstrap intervals
lowUpBound <- qnorm(0.95)*sd
lowerN <- round(estimate-lowUpBound,3)
upperN <- round(estimate+lowUpBound,3)
data.frame(normal = c(lowerN,upperN),
           percentile = round(quantile(thetaest1B,probs=c(0.05,0.95)),3),
           row.names = c("lower","upper"))
```

```
##       normal percentile
## lower  0.315      0.361
## upper  0.647      0.686
```

Comment: We see that the percentile interval is shifted to the right compared with the normal interval. We can see from the histogram, that the bootstrap samples are not normal. Thus the difference could be expected and the percentile interval is preferable in this situation.

Problem 5

The occurrence of earthquakes can be modeled as a homogeneous Poisson process with intensity parameter λ where λ is the expected number of earthquake per year. Let us assume we live in a region with $\lambda = 10$. As we have a homogenous Poisson process, the time between earthquakes $T_1, \dots, T_n$ follows an exponential distribution. Let us assume $T_1, \dots, T_n \sim Exp(a \cdot b)$, i.e. the rate parameter $\lambda = a \cdot b$ and $E(T_i) = \frac{1}{a \cdot b}$ with $a > 0$ and $b > 0$. Further, we have recorded $n = 20$ observations $t_i, i = 1, \dots, n$

$$0.08,\ 0.12,\ 0.01,\ 0.01,\ 0.04,\ 0.29,\ 0.12,\ 0.05,\ 0.10,\ 0.01$$
$$0.14,\ 0.08,\ 0.12,\ 0.44,\ 0.11,\ 0.10,\ 0.19,\ 0.07,\ 0.03,\ 0.06$$

and we assume the prior is given by $a \sim Exp(3)$ and $b \sim Exp(3)$.

a)

Given the density function $f(t) = \lambda e^{-\lambda t}$ of an exponential distribution, the Likelihood can be written as

$$L(\lambda) = \prod_{i=1}^{n} f(t_i; a, b) = \prod_{i=1}^{n} a \cdot b \cdot e^{-abt_i} = (ab)^n e^{-ab\sum_{i=1}^{n} t_i}$$

and the prior

$$p(\lambda) = p(a) \cdot p(b) = 3e^{-3a} \cdot 3e^{-3b} = 9e^{-3a-3b}.$$

Thus the posterior is

$$\begin{aligned} p(\lambda; t_1, \ldots, t_n) &\propto L(\lambda) \cdot p(\lambda) = (ab)^n e^{-ab\sum_{i=1}^{n} t_i} \cdot 9e^{-3a-3b} \\ &\propto (ab)^n e^{-ab\sum_{i=1}^{n} t_i} e^{-3(a+b)}. \end{aligned}$$

b)

We have

$$\begin{aligned} \log p(\lambda; t_1, \ldots, t_n) &\propto n\log(ab) - ab\sum_{i=1}^{n} t_i - 3(a+b) \\ &= n\log(a) + n\log(b) - ab\sum_{i=1}^{n} t_i - 3a - 3b \end{aligned}$$

and thus

$$\begin{aligned} \log p(a|b, t_1, \ldots, t_n) &\propto n\log(a) - ab\sum_{i=1}^{n} t_i - 3a \\ &= n\log(a) + (-b\sum_{i=1}^{n} t_i - 3) \cdot a \end{aligned}$$

Given that we know if $\log p(x) = a + b\log(x) + cx$ for $x > 0$ then $x \sim Gamma(b+1, -1/c)$ we have for the conditional posteriors

$$a|b, t_1, \ldots, t_n \sim Gamma\left(n+1, \frac{1}{b\sum_{i=1}^{n} t_i + 3}\right)$$

As a and b have the same prior, it follows

$$b|a, t_1, \ldots, t_n \sim Gamma\left(n+1, \frac{1}{a\sum_{i=1}^{n} t_i + 3}\right).$$

c)

This is Gibbs sampler, since the elements of the vector θ are updated according to the conditional distributions $p(a|b, t_1, \ldots, t_n)$ and $p(b|a, t_1, \ldots, t_n)$.

d)

This is a RWMH sampler with $N(a_{t-1}, 1.5^2)$ as the proposal distribution for a_t and $N(b_{t-1}, 1.5^2)$ as the proposal distribution for b_t.

e)

The Gibbs sampler is to be preferred in this case, because we are able to find the conditional proposal for each block and thus have a efficient sampling algorithm. For RWMH, we use normal distribution as a proposal. First, if we compare the two plots, we can observe that the proposal (normal distribution) has problems to explore the whole support of the target distribution. Second, as $a, b > 0$, we have to reject if $a^* \leq 0$ or $b^* \leq 0$.Thus, the RWMH sampler is less efficient than the Gibbs sampler.